



Enabling Onboard Navigation for UAVs in GPS Denied Environments

MIP400A : B.Tech Project - MidTerm Evaluation

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Goal: UAV self-localization when GPS/GNSS is unavailable, jammed, or spoofed, using only onboard sensors.

- ❑ **This semester:** working end-to-end pipeline, in simulation.
- ❑ **Next semester (through Apr 2027):** fast, power-efficient (FPGA) onboard deployment.

Core approach: Scene-Aided Navigation

Replace GPS with passive measurements the aircraft can already sense: terrain, gravity, magnetic field, and visual scene.



Weak signal

GNSS is easily jammed, spoofed, or simply unavailable.

Mechanical view

The IMU already senses acceleration, rotation, and gravity; a camera could replace the satellite link entirely.

Must keep flying

Survey, disaster-response, and defense UAVs need to operate even when GPS fails.

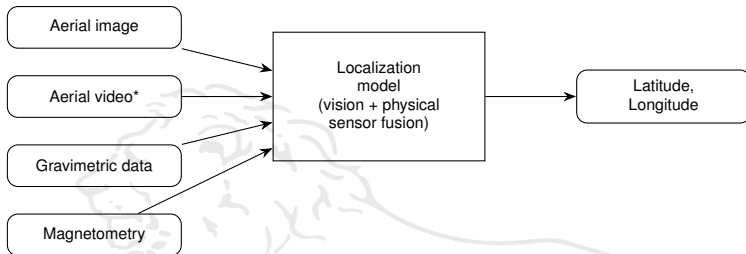
Precedent

TERCOM and DSMAC proved passive, GPS-free terrain and scene matching decades ago.

Passive GPS-denied navigation methods, screened for feasibility on a small UAV:

Method	Physical quantity measured	Passive?	Assessment
Terrain matching (TERCOM)	Ground elevation	Yes	Proven, strong candidate ^[4,6]
Visual landmark matching	Camera imagery	Yes	Strong if paired with AI matching ^[1-3,5,7,12,13]
Transformer-based matching	Camera imagery	Yes	SOTA in this niche; motivates our redesign ^[14,15]
Gravity matching	Gravity anomalies	Yes	Promising, needs sensitive sensor ^[8,9]
Gravity-gradient nav.	$\partial g / \partial x, y, z$	Yes	Very promising, higher precision ^[8,9]
Magnetic anomaly nav.	Magnetic field	Yes	Useful in mountainous terrain ^[10,11]
Quantum inertial sensing	Acceleration / rotation	Yes	No map needed; pure mechanics
Wi-Fi / cellular / RF	Radio signal strength	No	Rejected: unavailable at altitude

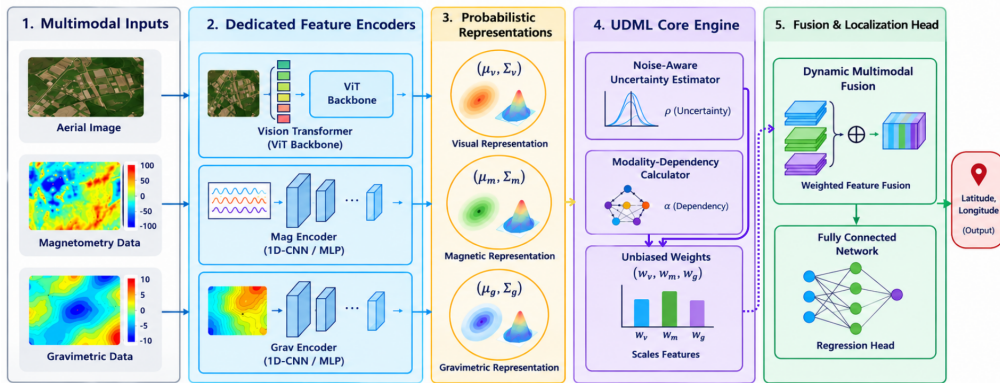
Reviewed: VGG16^[1], CLIP^[2,3,7], SIFT/ORB^[5], and DEM terrain-weighting^[4,6], achieving sub-3–7 m RMSE^[2,4].



*planned as a stream of individual images from the video stream sent to model to get similar results to image input but with a sense of direction as well

- ❑ **Baseline validated:** replicated the reference implementation^[1] in simulation; accuracy matches the paper's reported numbers.
- ❑ **Now:** redesigning the backbone as a **transformer** (next slide), trading compute and latency for higher accuracy, then adding multimodal sensor fusion.

Proposed Model Architecture



- Transformer vision backbone (ViT) replaces VGG16^[1]; magnetometry and gravimetry get their own 1D-CNN/MLP encoders.
- Each modality is encoded as a **probabilistic representation** (μ, Σ), so sensor noise is carried forward explicitly.
- The **UDML core** estimates per-modality uncertainty ρ and dependency α , producing unbiased fusion weights (w_v, w_m, w_g).
- Weighted features are fused and regressed to **latitude, longitude** by a fully connected head.

Multimodal sensor fusion needs a dataset with synchronized, geo-tagged measurements per sample:

Modality	Role
Aerial / satellite image	Visual scene input to the ViT backbone
Latitude, Longitude	Ground-truth label for localization
Magnetometer reading	Input to the magnetic-anomaly encoder
Gravimetric reading	Input to the gravity-anomaly encoder
Altitude	Scale normalization for image and physical-field readings

Challenge

No public dataset combines all five fields together; our advisor is helping procure aligned imagery with gravimetric/magnetometric survey data through IITR's Department of Earth Sciences (Geophysics).



Target	Milestone
Early Oct 2026	ViT vision backbone: initial training & testing complete (vision-only pipeline)
End Oct 2026	At least one physical modality (gravimetric <i>or</i> magnetometric) fused into the model
Nov 2026	Full multimodal fusion (vision + magnetometer + gravimeter), end-to-end pipeline in simulation
Jan – Apr 2027	Optimize for speed & power on FPGA; physical UAV test

- ❑ **Compute:** training and testing will run on rented cloud GPUs (e.g., Modal) or on-campus GPUs, depending on availability.
- ❑ Phase 2 (FPGA) extends our lab project from last semester, which showed FPGAs are highly power-efficient for this class of computation.

- [1] H. Goforth and S. Lucey, *GPS-Denied UAV Localization using Pre-existing Satellite Imagery*, CMU Robotics Institute. <https://publications.rti.cmu.edu/gps-denied-uav-localization-using-pre-existing-satelliteimagery> (code: <https://github.com/hmgoforth/gps-denied-uav-localization>)
- [2] *ngps_flight*: real-time CLIP-style UAV-satellite matching, ~3 m RMSE on Jetson Orin. https://github.com/snktshrma/ngps_flight
- [3] *CLIP-UAV-localization*: CLIP-based network matching UAV imagery, also recovers heading. <https://github.com/codebra721/CLIP-UAV-localization>
- [4] *GNSS-Denied-UAV-Geolocalization*: terrain-weighted optimization using satellite maps and DEM data, sub-7 m. <https://github.com/YFS90/GNSS-Denied-UAV-Geolocalization>
- [5] *VisionUAV-Navigation*: classical SIFT/ORB hybrid feature detection, no deep learning. <https://github.com/sidharthmohannair/VisionUAV-Navigation>
- [6] GNSS-denied UAV geolocalization via terrain-weighted constraint optimization, sub-7 m error, *ScienceDirect*. <https://www.sciencedirect.com/science/article/pii/S1569843224006332>
- [7] CLIP-based UAV-satellite image matching, also derives heading direction, *ScienceDirect*. <https://www.sciencedirect.com/science/article/pii/S2590123025041787>
- [8] A Review of Key Technologies in Gravity Matching Navigation, *Sensors* (2026). <https://www.mdpi.com/1424-8220/26/13/4208>
- [9] Gravity-Aided Navigation Using Viterbi Map Matching Algorithm. <https://arxiv.org/abs/2204.10492>
- [10] Absolute Positioning Using the Earth's Magnetic Anomaly Field. <https://doi.org/10.1002/navi.138>
- [11] MagNav-Based Navigation Simulation Using Magnetic Anomaly Maps for GPS-Denied Environments. <https://doi.org/10.5302/J.ICROS.2026.25.0344>
- [12] *PCVLabDrone2021*: real-time down-facing camera pipeline with quad-tree map retrieval, tested live. <https://github.com/OSUPCVLab/PCVLabDrone2021>
- [13] *geolocalization*: dual/Siamese network recognizing same-scene UAV and satellite image pairs. <https://github.com/abhinavtripathi95/geolocalization>
- [14] X. Zhu et al., *TransGeo: Transformer Is All You Need for Cross-view Image Geo-localization*, CVPR 2022. <https://arxiv.org/abs/2204.00097>
- [15] *FSRA*: A Transformer-Based Feature Segmentation and Region Alignment Method for UAV-View Geo-Localization. <https://arxiv.org/pdf/2201.09206>